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Paleomagnetism From Central Iran Reveals Arabia-Eurasia Collision Onset at the Eocene/Oligocene Boundary

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Key Points:

- Our paleomagnetic results indicate a ~3.7°–3° of latitude south of the present position of Central Iran during ca. 40–23 Ma
- Central Iran has experienced ~11.6° clockwise rotation since ca. 40 Ma
- Arabia-Eurasia collision began at the Eocene/Oligocene boundary in the western Main Zagros suture and diachronously spread eastwards

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract We present two robust and well-dated paleomagnetic poles from upper Eocene and Oligocene volcanics in the Urumieh-Dokhtar magmatic arc, Central Iran. These two poles place Iran ~3.7°–3° of latitude south of its present position between ca. 40 and 23 Ma. Our new paleomagnetic declination data indicate that the Central Iran block may have experienced a ~11.6° clockwise rotation since the Late Eocene. We integrated our new data with the retrodeformed margins of the Zagros collision zone and contemporaneous Arabia positions to better constrain the age and configuration of the Arabia and Eurasia assembly process. In our model, the Arabia-Eurasia collision occurred first in the western Main Zagros suture between ca. 35 and 30 Ma and then diachronously spread eastwards. Our paleogeographic reconstruction and initial continental collision timing supports the Arabia-Eurasia collision as a first-order driver of global cooling, Red Sea rifting, and Mediterranean extension.

Plain Language Summary The demise of the Neo-Tethyan ocean and accompanied continent-continent collisions created the thick crust and the low relief surfaces of the Iran Plateau and Tibetan Plateau. The onset timing and configuration in the Zagros collisional belt are critical for understanding the uplift of the Iran Plateau, tectonic evolution of the Mediterranean and Zagros regions, as well as the associated Cenozoic climate change. However, the age and configuration of the Arabia-Eurasia continental collision are hotly debated. Previous works generated competing collision timing estimates ranging from Late Cretaceous to Pliocene, with most estimates from Eocene to Miocene. By conducting geochronology and paleomagnetism on the Eocene-Oligocene volcanic rocks in Central Iran, we show that the Arabia-Eurasia collision occurred first in the western Main Zagros suture at the Eocene/Oligocene boundary, and then diachronously spread eastwards. We suggest the Arabia-Eurasia collision facilitates the slowing of Africa, the opening of the Red Sea, the extension in the Mediterranean, and the Eocene/Oligocene global cooling.

1. Introduction

Several profound tectonic events occurred around the Alpine-Zagros orogen during the Early Cenozoic, such as the ~25 Ma slowing of the Arabia-Eurasia convergence (e.g., McQuarrie et al., 2003), the Paleogene extensional arc flare-up in the Iran block (e.g., Verdel et al., 2011), ~40–30 Ma Afar plume upwelling and Red Sea rifting (e.g., Bosworth & Stockli, 2016; Faccenna et al., 2013), and ~30–25 Ma Mediterranean extension (e.g., Jolivet & Faccenna, 2000; van Hinsbergen et al., 2020). These major tectonic events and Eocene/Oligocene boundary global cooling (e.g., Coxall et al., 2005) have generally been interpreted to have had a positive or negative relationship with the Arabia-Eurasia continental collision based on the timing of the collision (e.g., Allen & Armstrong, 2008). Although much effort has been made to constrain the timing of the collision between the Arabian and Eurasian continents, estimates range widely from the late Cretaceous to the Pliocene, with most estimates between ca. 35 and 20 Ma (e.g., Hatzfeld & Molnar, 2010; McQuarrie and van Hinsbergen, 2013). The initial continent-continent collision and subsequent closing of the Tethyan seaway during Late Eocene-Early Oligocene time may have triggered these above contemporaneous tectonic events and global cooling (e.g., Allen & Armstrong, 2008; Ballato et al., 2011; Barber et al., 2018; Homke et al., 2010; Jolivet & Faccenna, 2000; Vincent et al., 2007). However, a younger collision age with estimates from late Oligocene to Pliocene would rule out this collisional mechanism (e.g., Fakhari et al., 2008; McQuarrie and van Hinsbergen, 2013; Mouthereau et al., 2007; Okay et al., 2010; Sun et al., 2021; Zhang et al., 2017).

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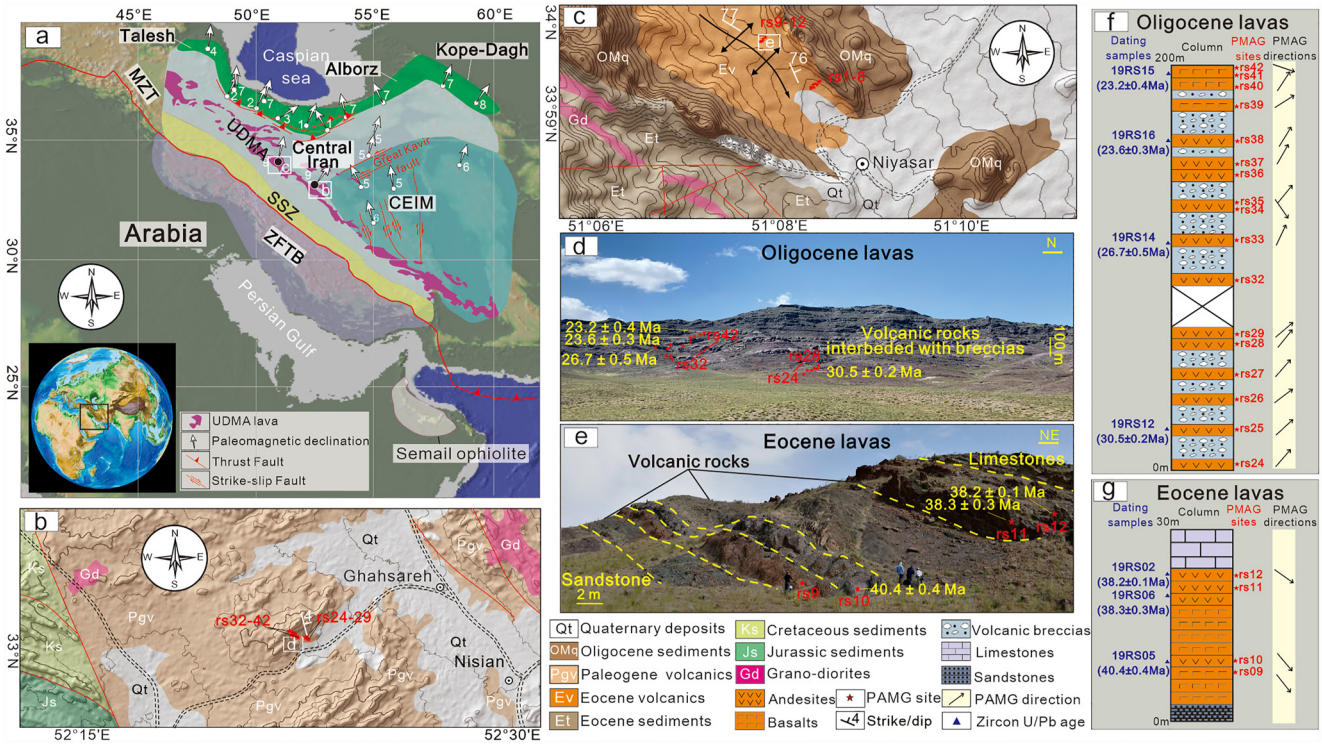


Figure 1. (a) Tectonic and topographic map of the Arabia-Eurasia collision zone, modified from Tadayon et al. (2017). White dots show the locations of paleomagnetic studies on Eocene-Miocene rocks in the Iran blocks. Black dots show the locations of our sampling areas. Arrows indicate the directions of paleomagnetic declinations: 1-Cifelli et al. (2015); 2-van der Boon et al. (2018); 3-Bina et al. (1986); 4-Mammedov and Nazarov (1971); 5-Mattei et al. (2012); 6-Mattei et al. (2020); 7- Mattei et al. (2017); 8-Mattei et al. (2019); 9-this study. Simplified geological maps and photographs of our sampled Oligocene lavas in the Marbin area (b, d) and Eocene lavas in the Kashan area (c, e). The column, samples, paleomagnetic sites, and paleomagnetic directions of our sampled Oligocene (f) and Eocene (g) lavas. Abbreviate: ZFTB-Zagros fold-thrust belt; MZT-Main Zagros thrust; SSZ-Sanandaj-Sirjan zone; UDMA-Urumieh-Dokhtar magmatic arc; CEIM-Central East Iranian Microcontinent; PMAG-Paleomagnetism.

The plate kinematic reconstructions and presence of Eurasian detritus in the Arabian foreland basin is direct evidence that can constrain the continent-continent collision age (e.g., Cai et al., 2021; Ding et al., 2017; GholamiZadeh et al., 2021; Koshnaw et al., 2019; Pirouz, Avouac, Hassanzadeh, et al., 2017). Eocene-Miocene paleomagnetic data from the deformed Iran orogen are required to place the orogen within the context of the global plate circuit. Available Cenozoic paleomagnetic data are still limited from the Iran plateau. In the northern Iran, few paleomagnetic studies have been conducted on Eocene-Miocene sediments (Figure 1a; Cifelli et al., 2015; Mammedov & Nazarov, 1971; Mattei et al., 2017, 2019) and volcanic rocks (van der Boon et al., 2018) of the Talesh-Alborz-Kope-Dagh Mts., to constrain the far-field rotation and convergence accommodated during the Arabia-Eurasia collision. Oligocene-Miocene paleomagnetism of the red beds of the Central East Iranian Microcontinent (CEIM) reveals opposite vertical-axis rotations associated with the different orientations and sense of movements of the left-lateral Great Kavir strike fault systems (Figure 1a; Mattei et al., 2012, 2020). However, the only paleomagnetic study from Eocene-Miocene volcanic rocks in the Central Iran block, which directly contacts with the Arabia plate, did not provide detailed demagnetization data, field tests, or a sufficient number of samples (less than six sites per locality) and might not have performed adequate demagnetizations with an alternating-field treatment on specimens containing hematite (Bina et al., 1986). Here, we present two well-dated and robust paleomagnetic poles from upper Eocene and Oligocene lavas to elucidate the latitudinal motions and rotations of the Central Iran block and then discuss the constraints on the timing and configuration of the Arabia-Eurasia collision by combining documented shortening, crustal subduction along the Zagros orogen, and coeval Arabian positions.

2. Geological Setting and Sampling

The Zagros orogen mainly consists of the Zagros fold-thrust belt (ZFTB), the Sanandaj-Sirjan zone (SSZ), and the Urumieh-Dokhtar magmatic arc (UDMA), from south to north (e.g., Barber et al., 2018; Figure 1). The Main

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Zagros Thrust (MZT), separating the ZFTB and SSZ, is widely proposed to represent the boundary between the Eurasian and Arabian plates, based on lines of evidence derived from ophiolites, seismic profiles, and metamorphism (e.g., Agard et al., 2011; Hatzfeld & Molnar, 2010). The UDMA displays a NW-SE linear magmatic belt, comprising voluminous volcanic successions and minor intrusive rocks. Although the UDMA has recorded subduction-related volcanism since the Jurassic, a notable magmatic flare-up occurred during the Eocene to Miocene (55–15 Ma) (e.g., Chiu et al., 2013; Nouri et al., 2021). We carried out a combined geochronological and paleomagnetic study on the Eocene-Oligocene lavas that crop out in the central part of the UDMA (Figure 1), where previous detailed structural investigations suggest that large-scale folds and uplift of the UDMA massifs obviously occurred during the late Miocene-Pliocene (Morley et al., 2009). We collected a total of 323 paleomagnetic core samples from 30 volcanic sites and seven U/Pb dating samples from two localities: (a) the Kashan locality (33°59′8.61″N, 51°8′33.7″E), upper Eocene lavas, two sections, 13 paleomagnetic sites and three zircon U/Pb dating samples; (b) the Marbin locality (33°00′14.91″N, 52°22′7.51″E), Oligocene lavas, one section, 17 paleomagnetic sites and four zircon U/Pb dating samples (Figure S1 in Supporting Information S1). At least seven Eocene volcanic layers in the sampled Kashan locality are conformably overlain with limestones (Figure 1g). More than 14 Oligocene volcanic layers drilled in the Marbin locality are conformably interbedded with volcanic breccia (Figure 1f). The sampled volcanic rocks in the Kashan and Marbin localities mainly comprise andesites, basaltic andesites, and basalts. See Supporting Information S1 for details about the sampling strategy, procedures, and descriptions of the methods and the data for U/Pb dating, rock magnetism analysis, magnetic petrography, and paleomagnetism analysis.

3. Results

3.1. Geochronology and Paleomagnetism of the Eocene Lavas

Our new zircon U/Pb LA-ICP-MS analyses of three andesites/basaltic andesites samples from the lower and upper parts of section B (sites rs09-12) in the Kashan provide concordant $^{206}Pb/^{238}U$ mean ages of 40.4 ± 0.4 Ma (MSWD = 1.9, n = 12), 38.3 ± 0.3 Ma (MSWD = 0.46, n = 29), and 38.2 ± 0.1 Ma (MSWD = 0.004, n = 37) (Figure S2 and Table S1 in Supporting Information S1). Considering the consistent rock types and relations with overlain limestones between section A (sites rs01-08) and section B (Figure S3 in Supporting Information S1), we believe the age of our sampled volcanic rocks in both sections is Late Eocene (ca. 38–40 Ma).

Two well-defined magnetic components were successfully isolated from most paleomagnetic specimens by stepwise thermal demagnetization (TD) (Figure 2a). The mean direction of the low-temperature component (LTC) generally determined below 350°C, which yields a negative fold test, is probably a viscous remanent magnetization acquired recently (Figure 2b). The high-temperature component (HTC) is generally defined from ~350°C up to 600°C or 680°C, and is carried by magnetite and hematite (Figure 2; Figures S4–S6 in Supporting Information S1). We thus calculate a mean pole at 20.3°N, 344.2°E (A_{95} = 19.3°, K = 6.6, S = 32.4) in in situ coordinates by averaging 11 site-mean virtual geomagnetic poles (VGPs). The Fisher mean VGP in tilt-corrected coordinates is located at 80.2°N, 160.4°E (A_{95} = 9.5°, K = 24.0, S = 16.9), resulting in a tilt-corrected mean direction of 49.4 ± 8.8° inclination and 10.7 ± 8.6° declination (Tables S2 and S3 in Supporting Information S1).

Eleven site-mean directions of the upper Eocene lavas passed a fold test, with a high degree of confidence (i.e., $\xi 2$ = 9.16 before, and 3.68 after, tilt correction; critical value ξ = 3.87 at the 99% confidence level and 5.38 at the 95% confidence level) (McFadden, 1990). A progressive unfolding correction shows that the site-mean directions exhibit the highest precision parameter at 111.7% unfolding (with 95% uncertainties ranging from 103.6% to 119.9%) (Watson & Enkin, 1993), with k increasing largely from 8.9 (in situ) to 31.2 (tilt corrected), and after tilt correction, the HTC directions from the two sections share a common true mean direction. Thus, we conclude that the 11 site-mean directions presented here record a pre-folding magnetization. Structural investigations confirm a late Miocene age of the substantial phase of folding of UDMA (Morley et al., 2009). This combined with no obvious observations of alteration from rock magnetism and petrography, strongly support an interpretation of a primary thermoremanent origin (see details in Supporting Information S1). The S [angular dispersion of poles] = 16.9° value of our sampled Eocene lavas is consistent with that calculated from the PSV10 data set at a similar latitude (Cromwell et al., 2018; de Oliveira et al., 2021). The radius of the VGP confidence cone (A_{95} = 9.5°) falls well within the N-dependent reliability interval (6.5–18.1°) expected for the PSV of a population of 11 poles (Deenen et al., 2011). Our analysis supports that these results most likely do characterize the time-averaged behavior of the geomagnetic field.

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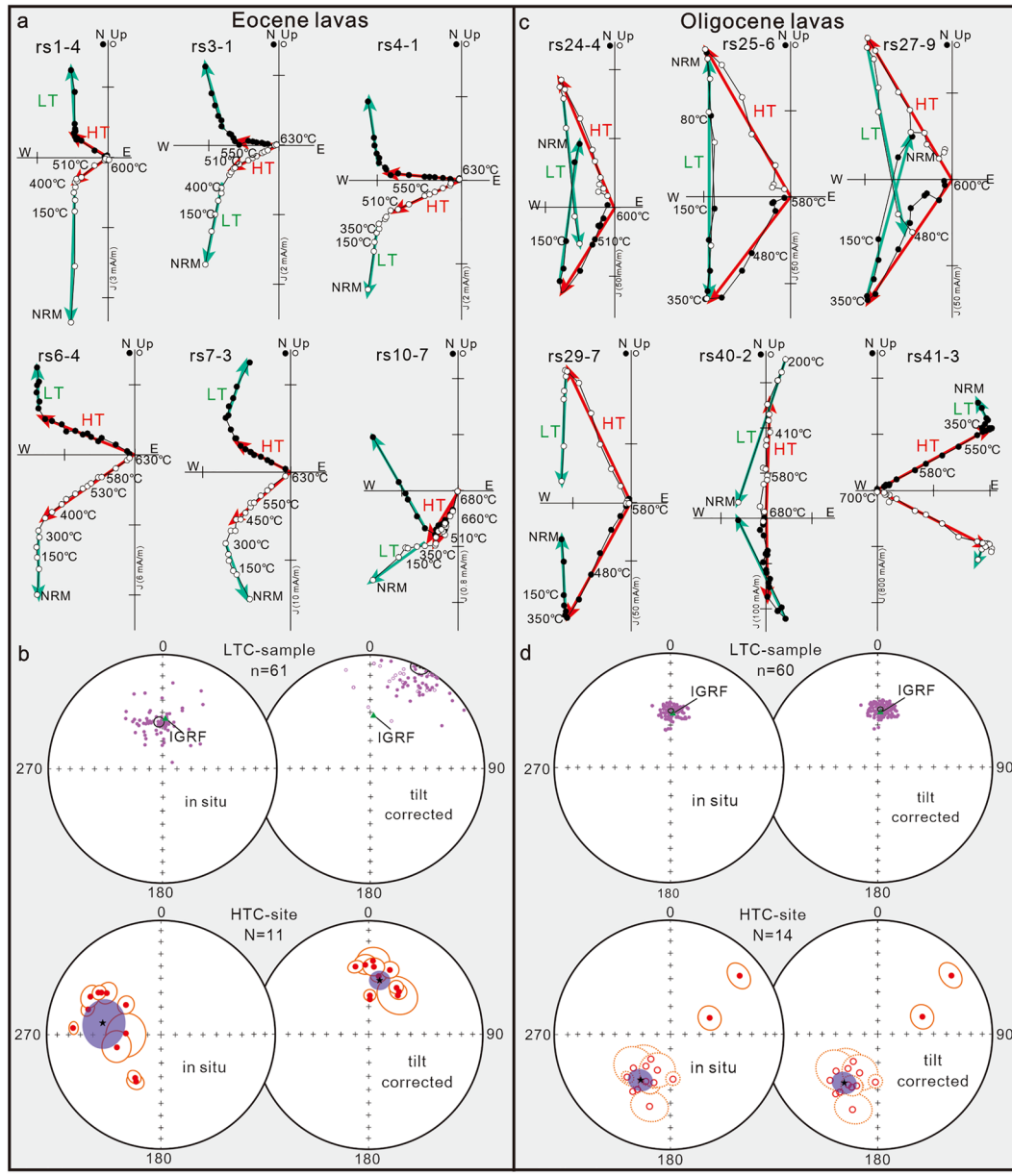


Figure 2. Orthogonal vector diagrams of the thermal demagnetization behaviors of typical Eocene (a) and Oligocene (c) volcanic specimens in in situ coordinates, with views of the low-temperature (LT) and high-temperature (HT) components. Equal-area projections of the paleomagnetic sample-mean low-temperature component (LTC) and site-mean high-temperature component (HTC) directions of the Eocene (b) and Oligocene (d) lavas.

3.2. Geochronology and Paleomagnetism of the Oligocene Lavas

Four andesites/basaltic andesites from the lower and upper parts were dated, yielding concordant $^{206}Pb/^{238}U$ mean ages of 30.5 ± 0.2 Ma (MSWD = 1.8, n = 33), 26.7 ± 0.2 Ma (MSWD = 0.5, n = 36), 23.6 ± 0.3 Ma (MSWD = 0.3, n = 26), and 23.2 ± 0.4 Ma (MSWD = 0.1, n = 10) (Figure S2 and Table S1 in Supporting Information S1). We assigned an Oligocene (~30.5–23.2 Ma) age as the crystallization age of our sampled section in Marbin.

The rock magnetic data suggest that hematite and magnetite are the magnetic remanence carriers (Figure S6 in Supporting Information S1). Two well-defined magnetic components were cleared in most specimens by TD treatments (Figure 2). The LTC and HTC were generally determined below 350°C and between 350°C and

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680°C, respectively. The LTCs, are probably viscous remanent magnetizations acquired recently (Figure 2). The HTCs yield both geomagnetic polarities, with more vectors directed southwest and up (12 sites) and fewer directed northeast and down (2 sites). The reversal test for the two polarities is indeterminate, which may be due to only two sites yielding different polarities. Nevertheless, the site-mean direction of the two reversal sites is antipodal to that of 12 sites showing normal polarities. We thus averaged the 14 site-mean VGPs to calculate a mean pole at 63.4°N, 131.9°E (A_{95} = 9.0°, K = 20.7, S = 18.0) in in situ coordinates. The Fisher mean VGP in tilt-corrected coordinates is located at 60.2°N, 136.9°E (A_{95} = 8.5°, K = 22.8, S = 17.1), yielding a mean direction of 50.1 ± 7.6° inclination and 35.2 ± 7.9° declination (Tables S2 and S4 in Supporting Information S1).

We note that there is no obvious observation of alteration based on our rock magnetism and petrographic observations (see the Figures S6–S8 in Supporting Information S1). This finding, combined with the presence of dual polarities, indicates that these 14 sites might record primary thermoremanent magnetizations. A reasonable estimate of paleosecular variation (S = 17.1 and A_{95} = 8.5 within the observed range of the modern geomagnetic field (5.9°–15.5°)) and dual polarities suggest secular variation should have been successfully averaged out and are suitable for paleographic reconstruction.

4. Discussion

To better constrain the age and configuration of the collision between the Arabian and Eurasian plates, we investigated the Eocene-Oligocene kinematic reconstructions of the Arabia and Eurasia collision zone. At first, we compiled the estimates of documented shortening and continental crust subduction within the Zagros orogen to reconstruct the previous locations of each continental edge. Then, we followed the Arabia-Nubia-North America-Eurasia plate circuit rotation method of McQuarrie and van Hinsbergen (2013) and integrated the compiled shortening estimate within the Zagros belt, to provide precise estimates of Arabian motions. The Eocene-Oligocene positions of Eurasia could be well constrained by our new and previous paleomagnetic data, as well as documented shortening. We limit our reconstruction of Eurasia to the Iran block, including the Talesh-Alborz-Kope-Dagh, Central Iran and CEIM blocks (Figure 3), which accreted to the southern margin of Eurasia in the latest Triassic (e.g., Muttoni et al., 2009).

Although the documented complex shortening of the basement and cover strata across the ZFTB began in the Late Cretaceous, the deformation manifesting as out-of-sequence and in-sequence thrusts and associated folds accelerated from 20 to 10 Ma and has continued to migrate southwards up to the present (e.g., Barber et al., 2018; Gavillot et al., 2010; McQuarrie, 2004; Mouthereau et al., 2012). Variable shortening estimates of the basement and cover, as constrained by balanced cross-sections, range from a maximum of 175 km to a minimum of 50 km (Alavi, 2007; Blanc et al., 2003; McQuarrie and van Hinsbergen, 2013; Mouthereau et al., 2007; Sherkati et al., 2006). Underthrusting of the Arabian continental crust is evidenced by the well-outcropped Semail ophiolite of Oman, which may have emplaced ∼180 km over the Arabian passive margin during the Late Cretaceous (e.g., Guilmette et al., 2018; Searle & Cox, 1999). The simple flexural model at the SSZ orogen scale suggests a ∼65 km of along-strike averaged crustal shortening for the southern margin of the Central Iran block (Pirouz, Avouac, Hassanzadeh, et al., 2017; Pirouz, Avouac, Gualandi, et al., 2017).

Previous Eocene-Miocene paleomagnetic declinations of the Talesh-Alborz-Kope-Dagh Mts. yield opposite vertical axis rotations along its strike (Figure 3d; Cifelli et al., 2015; Mammedov & Nazarov, 1971; Mattei et al., 2017, 2019; van der Boon et al., 2018). A secondary arc model was proposed to explain the kinematics inferred by these paleomagnetic data by Mattei et al. (2017) and van der Boon et al. (2018) as follows: the curved mountain belt was originated from a primary linear orientation, which was acquired after the middle-late Miocene. Therefore, we use a nearly E-W linear shape of the Talesh-Alborz-Kope-Dagh belt in our Eocene-Oligocene paleogeographic reconstructions. Compiled Oligocene-Miocene paleomagnetic results of CEIM also reveal opposite vertical-axis rotations (Mattei et al., 2012, 2020). Mattei et al. (2020) suggested these rotations were associated with the variable orientations and sense of movement of the Great Kavir strike fault systems, which experienced a shift from the Mesozoic-early Cenozoic right-lateral shearing to an active left-lateral motion during the latest Neogene (Mio-Pliocene) (Allen et al., 2011; Javadi et al., 2013, 2015; Tadayon et al., 2017; Zanchi et al., 2021). We compared our Eocene-Oligocene and previous Middle-Late Miocene observed declinations (Figure 1; this study; Mattei et al., 2012) from Central Iran with the present-day direction of the geocentric axial dipole field (GAD) for the reference point (34°N, 51.1°E) to discuss its rotation patterns (Figure S9 in Supporting Information S1). A significant declination difference is observed between Oligocene (∼35.2 ± 7.9°, Kashan) and

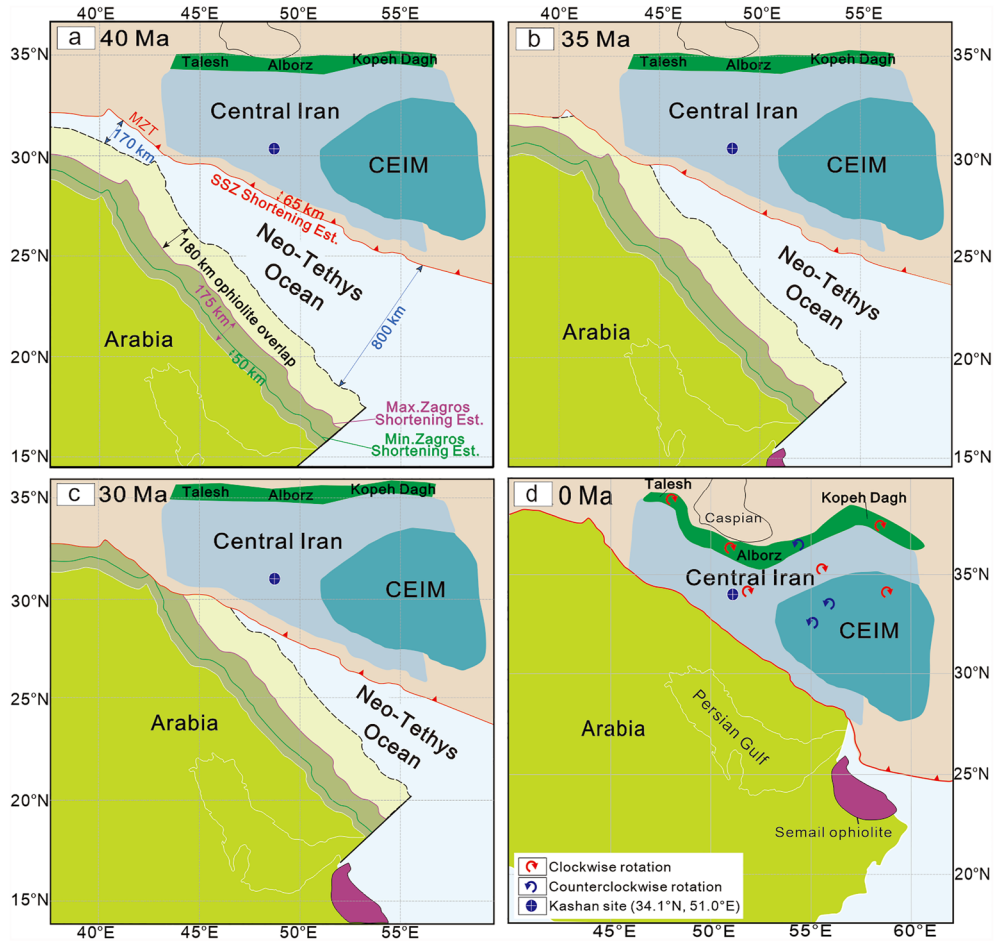


Figure 3. Paleogeographic reconstructions of the Arabia-Eurasia collision at 40, 35, 30, and 0 Ma in the rectangular coordinates. The positions of Iran at 40 and 35 Ma are based on our data from the Kashan lavas. The position of Iran at 30 Ma is based on our data from the Marbin lavas. The rotation models are based on the compiled paleomagnetic declinations in Figure 1. The positions of Arabia are based on McQuarrie and van Hinsbergen (2013). The blue circles show the location of our samples in the Kashan area (34°N, 51.1°E).

Eocene-Miocene (~10.7 ± 8.6°, Marbin area; ~11.4 ± 11.7°, Torud area; ~14.2 ± 13.9°, Jandaq area), which we suggest the Oligocene declination data might reflect a local rotation related to the right-lateral Kashan fault (Allen et al., 2011). No rotations of Central Iran appear to have occurred during the time of Late Eocene-Miocene. However, a ~11.6 ± 6.1° (averaged value of Eocene and Miocene declination data) post-Miocene clockwise rotation can be clearly inferred. Based on above, we made a ~11.6° counterclockwise rotation of the Central Iran and CEIM blocks to reconstruct their Eocene-Oligocene orientations, as we prefer a uniform post-Oligocene kinematic orientation between Central Iran and CEIM in our reconstructions. Our paleomagnetic data also show that the reference point (34°N, 51.1°E) for the Central Iran block was at latitudes of 30.3 ± 9.5°N and 31.0 ± 8.5°N during Late Eocene (40–38 Ma) and Oligocene (31–23 Ma), respectively. Although our paleomagnetic data cannot constrain the longitude, we can refer to the relative longitudinal constraints provided by previous plate circuits (McQuarrie and van Hinsbergen, 2013).

Our paleogeographic reconstruction suggests that at ca. 40 Ma, the restored front of the Arabian continent was still separated from that of Iran by at least ~170 km (northwest) to ~800 km (southeast) (Figure 3a). The largest estimate of the restored Arabian plate, extending an additional 355 km (including 175 km of Zagros shortening and 180 km of ophiolite obduction), should overlap with the restored Eurasian margin in the western portion of the MZT since ca. 37–35 Ma. Furthermore, the smallest estimate of the extension of Arabia (50 km) requires the collision with the restored Eurasian margin to have occurred at ca. 30 Ma (Figure 3). These paleogeographic reconstructions clearly suggest that the initial Arabia-Eurasia collision must have occurred in the western MZT by

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the latest Eocene-earliest Oligocene (Figure 3). Consistent with the retrodeformed configuration of the collision zone by McQuarrie and van Hinsbergen (2013), our paleogeographic model also suggests that the Arabia-Eurasia collision occurred diachronously. Arabia first collided with Eurasia in the western MZT at the Eocene/Oligocene boundary, and the collision then propagated eastwards.

This paleogeographic reconstruction for Arabia-Eurasia assembly is in general agreement with estimates based on collisional deformation (e.g., Barber et al., 2018; Homke et al., 2009) and observations of the foreland-basin sedimentation pattern (e.g., Cai et al., 2021; Koshnaw et al., 2019). Our paleomagnetic data, as well as the previous Eocene-Miocene paleomagnetic declinations of the Central Iran block (Mattei et al., 2012), suggest a clockwise rotation of the Iran block since the Eocene, consistent with the diachronous collision model (Figure 1a). This Eocene/Oligocene boundary Arabia-Eurasia collision would facilitates the collisional mechanism for the slowing of Africa, the opening of the Red Sea, the extension in the Mediterranean, and the Eocene/Oligocene global cooling.

5. Conclusions

We report two robust and well-dated paleomagnetic paleopoles from upper Eocene and Oligocene volcanics in the Urumieh-Dokhtar magmatic arc, Central Iran. These data average out secular variation and thus can be used for reconstruction. Our new results indicate that the Iran block was positioned at ∼30.3°–31.0°N during ca. 40–23 Ma. Our new paleomagnetic declination data indicate that the Central Iran block may have experienced a ∼11.6° clockwise rotation since the Late Eocene. Combined with the retrodeformed margins of the Zagros collision zone and contemporaneous Arabia positions, our results reveal that the Arabia-Eurasia collision occurred first in the western Main Zagros suture between ca. 35 and 30 Ma and then diachronously spread eastwards.

Data Availability Statement

Our data in this study are available for download through Figshare at https://doi.org/10.6084/m9.figshare.21274038.v2.

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